Complex Analysis

$z = x + iy$, ordered pair $(x, y) \equiv z \in \mathbb{C}$.
$= re^{i\theta}$ $\begin{cases} 0 \leq \theta \leq 2\pi \\ r > 0. \end{cases}$

$$z = x+iy \in \mathbb{C} \xrightarrow{f} f(z) = u(x,y) + iv(x,y) \in \mathbb{C}.$$

e.g., $f(z) = e^z$

$u(x,y) = e^x \cos y$

$v(x,y) = e^x \sin y$

$\lim_{z \to z_0} f(z) = w_0$ means

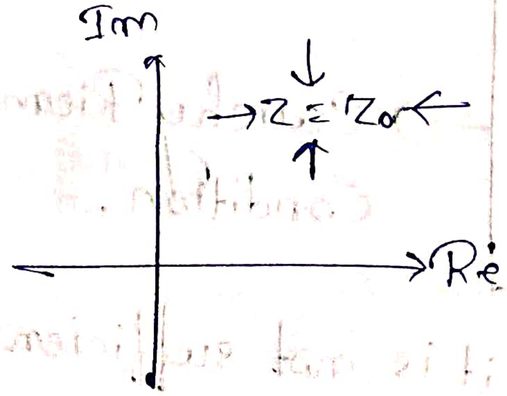
if for every +ve ε

$|f(z) - w_0| < \varepsilon$

we can find a +ve number δ such that

$0 < |z - z_0| < \delta$ for all z.

The limit w_0 should be same for all directions of approach to z_0.



Different from $f(x)$ real functions.

$f(z)$ is continuous at z_0 if it is defined at z_0 & $\lim_{z \to z_0} f(z) = f(z_0)$.

• $f(z)$ is differentiable at $z = z_0$ if

$$\lim_{z \to z_0} \frac{f(z) - f(z_0)}{(z - z_0)}$$ exists and we denote

it as $f'(z)$

$$f'(z) = \frac{u(x,y) - u(x_0,y_0) + i\,v(x,y) - v(x_0,y_0)}{(x - x_0) + i(y - y_0)}$$

Approach z_0 along parallel real axis

$$z = x + iy_0$$

$$= \lim_{x \to x_0} \frac{u(x,y_0) - u(x_0,y_0)}{(x - x_0)} + i\,\frac{v(x,y_0) - v(x_0,y_0)}{(x - x_0)}$$

$$= \frac{\partial u}{\partial x}(x_0,y_0) + i\,\frac{\partial v}{\partial x}(x_0,y_0).$$

If we approach along y-axis –

$$f'(z_0) = i\,\frac{\partial u}{\partial y}(x_0,y_0) + \frac{\partial v}{\partial y}(x_0,y_0)$$

• Necessary condition for differentiability

$$\boxed{\begin{aligned} \frac{\partial u}{\partial x} &= \frac{\partial v}{\partial y} \\ \frac{\partial u}{\partial y} &= -\frac{\partial v}{\partial x}. \end{aligned}} \longrightarrow \text{Cauchy Riemann Condition.}$$

↳ But it is not sufficient condition.

Example :-

1) $f(z) = x + iy$

$$\left.\frac{\partial u}{\partial x} = 1, \ \frac{\partial v}{\partial y} = 1, \ \frac{\partial u}{\partial y} = 0, \ \frac{\partial v}{\partial x} = 0\right\}$$

$\Rightarrow$ Possibly $f'(z)$ exists in whole of complex plane.

2) $f(z) = x - iy = z^*$

$$\left.\begin{array}{l} \frac{\partial u}{\partial x} = 1, \ \frac{\partial v}{\partial y} = -1 \\ \frac{\partial u}{\partial y} = 0 = \frac{\partial v}{\partial x} \end{array}\right\} \text{does not satisfy CR condition.}$$

$\Rightarrow f(z) = z^*$ is not differentiable in whole $\mathbb{C}$.

- Differentiability is a statement about $z = z_0$.

Example :-

Let $f(z) = \begin{cases} \frac{x^3}{x^2+y^2} + i\frac{y^3}{x^2+y^2}, & (x,y) \neq (0,0) \\ 0, & (x,y) = (0,0). \end{cases}$ $\rightarrow$ Is it differentiable at origin?

$$\Rightarrow \frac{\partial u}{\partial x} = \lim_{x\to 0} \frac{u(x,0) - u(0,0)}{x} = \lim_{x \to 0} \frac{\frac{x^3}{x^2} - 0}{x} = \lim_{x\to 0} \frac{x}{x} = 1$$

$$\frac{\partial v}{\partial y} = \lim_{y \to 0} \frac{v(0,y) - v(0,0)}{y} = \lim_{y\to 0} \frac{\frac{y^3}{y^2}}{y} = 1$$

$$\left.\frac{\partial u}{\partial x}\right|_{(0,0)} = \left.\frac{\partial v}{\partial y}\right|_{(0,0)} \rightarrow \text{1st CR condition satisfy.}$$

- Check 2^{nd} CR condition also satisfies at $(0,0)$ by explicitely checking the derivatives.

- $f(z) \rightarrow$ CR condition necessary.

$\rightarrow$ Does $f'(x=0, y=0)$ exists?

<u>Check</u>: $f'(0) = \lim_{z\to 0} \frac{f(z)-f(0)}{z-0}$

$= \lim_{z\to 0} \frac{f(z)}{z}$

Real axis $(y=0)$:- $f'(0) = \lim_{x\to 0} \frac{\frac{x^3}{x^2}}{x} = 1$

Imaginary axis $(x=0)$:-

$f'(0) = \lim_{y\to 0} \frac{i\frac{y^3}{y^2}}{iy} = 1$

If $y = x$,

$z = x + ix$

$f(x,x) = \frac{x^3}{x^2+x^2} + i\frac{x^3}{x^2+x^2}$

$= \frac{x}{2} + i\frac{x}{2}$.

$\lim_{z\to 0} \frac{\frac{x}{2}+i\frac{x}{2}}{x+ix} = \frac{1}{2} \neq 1$

- Example shows CR condition satisfies but $f(z)$ is not differentiable (at $z=(0,0)$.

$\rightarrow$ CR condition's not sufficient for differentiability.

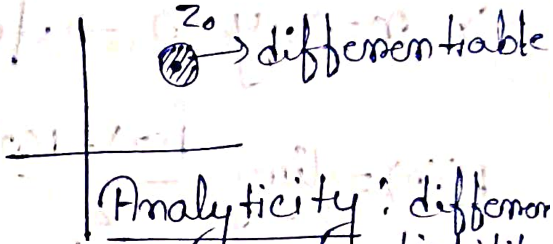
'Sufficient condition for differentiability :-

(i) CR conditions to be satisfied and.

(ii) ∂u, ∂v etc must exists be continuous at $z = z_0$. for $f'(z)$ to exist.

- Analytic Functions :-

A function $f(z)$ is analytic at $z = z_0$ of its domain of defination(D) if there exists a neighbourhood $|z - z_0| < \varepsilon\ (>0)$ such that $f'(z)$ exists in that neighbourhood.



Analyticity : differentiability in a region

- If CR is not satisfied ⇒ Non-Analytic.

Example :-

$f(z) = z^2$

$u = (x^2 - y^2)$; $v = 2xy$

$$\frac{\partial u}{\partial x} = 2x = \frac{\partial v}{\partial y} \; ; \; \frac{\partial u}{\partial y} = -2y = -\frac{\partial v}{\partial x}.$$

- CR conditions satisfied in whole $\mathbb{C}$.
- condition II : $\partial_x u, \partial_y v$ ... continuous.

→ Differentiable in whole $\mathbb{C}$.

⇒ This function is analytic.

Example :-

$f(z) = |z|^2 = x^2 + y^2$

$\Rightarrow u = x^2 + y^2, \; v = 0$

$\frac{\partial u}{\partial x} = 2x, \; \frac{\partial u}{\partial y} = 2y \;\Big|\; \frac{\partial v}{\partial x} = 0 = \frac{\partial v}{\partial y}$.

↳ Here CR is not satisfied in whole region except $(x,y) \equiv (0,0)$.

$x \neq 0, y \neq 0 \Rightarrow$ Not differentiable (in neighbourhood of $(0,0)$)
Not analytic.

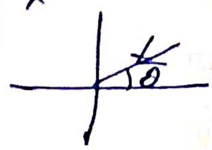
$\Rightarrow x = 0, y = 0 \rightarrow$ function is non-analytic

$$f'(0) = \lim_{z\to 0} \frac{f(z) - f(0)}{z}$$

$$= \lim_{z\to 0} \frac{|z|^2}{z} = \lim_{z\to 0} z^*$$

Take limit from any arbitrary ~~point~~ angle θ.

$z = re^{i\theta}$



$$\Rightarrow f'(0) = \lim_{z\to 0} z^* = \lim_{z\to 0} re^{-i\theta}$$

$$= \lim_{r\to 0} re^{-i\theta} = 0$$

for every value of θ.

$\Rightarrow f'(0) = 0$.

$f(z) = |z|^2$ non analytic in whole of $\mathbb{C}$.

↳ Non differentiable at $z \neq 0$

↳ Differentiable at $z = 0$

$f'(z=0) = 0$.

Complex Analysis

Recap :-

1) Concept of Limit :- must exist from all directions approaching a point and all must be same.

Necessary (but not sufficient) conditions :- Cauchy Riemann (C.R) conditions.

2) Differentiability is a concept about a point $z = z_0$.

3) Analyticity :- Differentiable in the open neighbourhood if $z = z_0$.

Ex :- $f(z) = |z|^2$

$\Rightarrow$ • $f(z)$ non analytic in whole complex plane.
• $f(z)$ differentiable at $z = 0$.

• If $f(z)$ is analytic at $z = z_0$; z_0 is a regular point. Otherwise it is 'Singular point'.

Isolated Singular Points :-

A Singular point that is surrounded by regular points.

Ex :- $f(z) = \frac{1}{z}$

$\Rightarrow z = 0$ singular (isolated), $z \neq 0 \Rightarrow$ analytic.

Ex :-

$x = \frac{z + z^*}{2}$, $y = \frac{z - z^*}{2i}$ $\longrightarrow$ better to write $g(z, z^*)$ because the function changes when we write it in terms of z and z^*.

$\Rightarrow f(x, y) = f(z, z^*) = u + iv$

Now, $\frac{\partial f}{\partial z^*} = \frac{\partial f}{\partial x}\frac{\partial x}{\partial z^*} + \frac{\partial f}{\partial y}\frac{\partial y}{\partial z^*}$

$= \frac{1}{2}\frac{\partial f}{\partial x} + \frac{i}{2}\frac{\partial f}{\partial y}$

$= \frac{1}{2}\left(\frac{\partial u}{\partial x} + i\frac{\partial v}{\partial x}\right) + \frac{i}{2}\left(\frac{\partial u}{\partial y} + i\frac{\partial v}{\partial y}\right) = 0$ [from CR condition

Alternate :- CR condition should satisfy $\frac{\partial f}{\partial z^*} = 0$

Note :- There should ~~be not~~ be not be any explicitely dependency on z^*.

Ex:- $f(x) = |z|^2 = zz^*$

$\Rightarrow f(x,y) = x+2iy = g(z, z^*)$

$\frac{\partial f}{\partial z^*} \neq 0 \Rightarrow$ non analytic function.

CR Condition :-

$$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y} \quad \Big| \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$$

$$\Rightarrow \frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0 \quad \Big| \quad \Rightarrow \frac{\partial^2 v}{\partial y^2} + \frac{\partial^2 v}{\partial x^2} = 0$$

$\Rightarrow$ v and u satisfy Laplace equation.

If f is analytic and g is analytic

1) $f \pm g$ is also analytic
2) fg is also analytic
3) $\frac{f}{g}$ is analytic in the domain where $g \neq 0$

Def^n (Entire Function) :-

If ~~the~~ a function is analytic in the whole of the complex plane, it is called entire function.

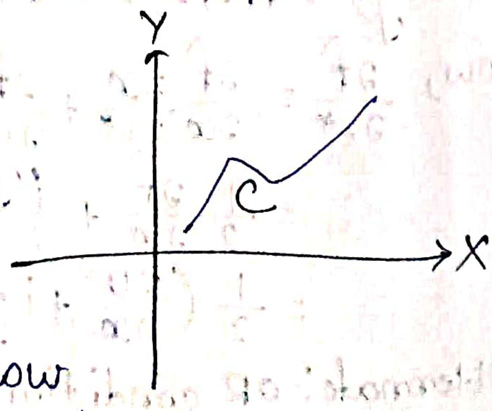
Ex:- $f(z) = e^z$, $f(z) = z$.

Contour Integration

Contour :- piecewise regular curve in complex plane.

Parametric form :-

$z(t) = x(t) + iy(t)$, t is real parameter.



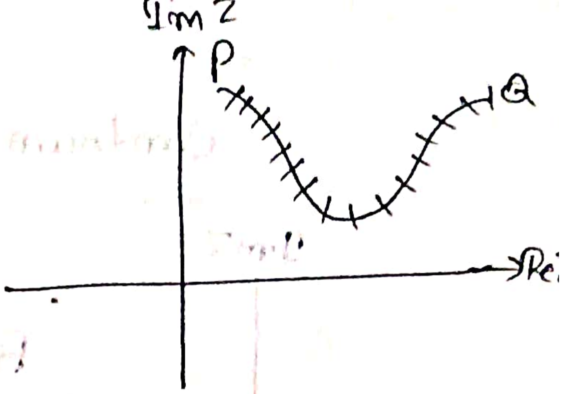
$I = \int_C f(z)\, dz$.

C is the path which we will follow for the integration. C is called contour.

$P = z_0, z_1, z_2, \ldots, z_n = Q$

Intermediate points $\alpha_1, \alpha_2, \ldots$

$\Delta z_j = z_j - z_{j-1}$



$$I = \int_C f(z)\,dz = \sum_{j=1}^{n} f(\alpha_j)\,\Delta z_j$$

Plug, $z = x + iy \Rightarrow f(z) = u + iv$.

$z_j = x_j + iy_j$, $\Delta z_j = \Delta x_j + i\Delta y_j$

$\alpha_j = \xi_j + i\eta_j$

$$I = \sum_{j=1}^{n} \left[u(\xi_j, \eta_j) + iv(\xi_j, \eta_j)\right]\left[\Delta x_j + i\Delta y_j\right].$$

$$= \sum \Big\{ \left[u(\xi_j, \eta_j)\Delta x_j - v(\xi_j, \eta_j)\Delta y_j\right] + i\left[v(\xi_j, \eta_j)\Delta x_j + u(\xi_j, \eta_j)\Delta y_j\right] \Big\}.$$

$$= \int_C \left[(u\,dx - v\,dy) + i(v\,dx + u\,dy)\right].$$

$$\Rightarrow I = \int_C f(z)\,dz = \int_C (u + iv)(dx + i\,dy) = \int (u\,dx - v\,dy) + i\int (v\,dx + u\,dy)$$

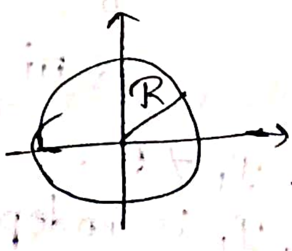
$$\Rightarrow I = \int (u\,dx - v\,dy) + i\int (v\,dx + u\,dy).$$

Ex :-

$I = \oint \frac{1}{z}\,dz$, over a circle of radius R $(R \neq 0)$.

$\Rightarrow z = 0 \rightarrow$ bad (non analytic)

On the path $\rightarrow$ No issue.



Parametric form

$z(t) = Re^{i2\pi t}$.

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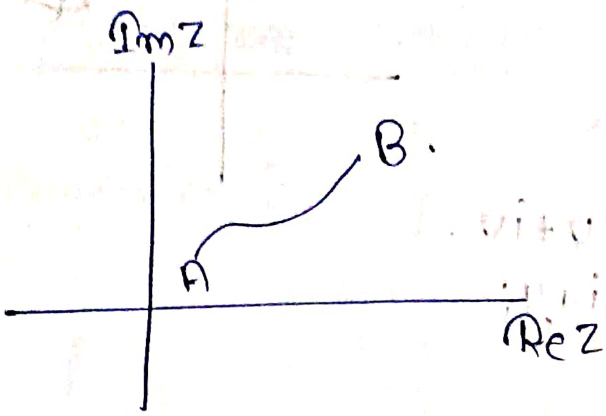
Contour Integrals



$$\int_C f(z)\,dz = \int (u\,dx - v\,dy) + i(v\,dx + u\,dy)$$

$$= \int_{t_1}^{t_2} f(z(t))\,\frac{dz}{dt}\,dt$$

Ex-1:-

$\oint \frac{1}{z}\,dz$, along a circle of radius R.

$\Rightarrow$ $z(t) = Re^{i2\pi t}$, $0 \le t \le 1$.

$$I_1 = \oint \frac{1}{z}\,dz$$

$$= \int \frac{1}{z}\frac{dz}{dt}\,dt$$

$$= \int_0^1 R^{-1} e^{-2\pi i t} (R\,2\pi i)\, e^{i2\pi t}\,dt$$

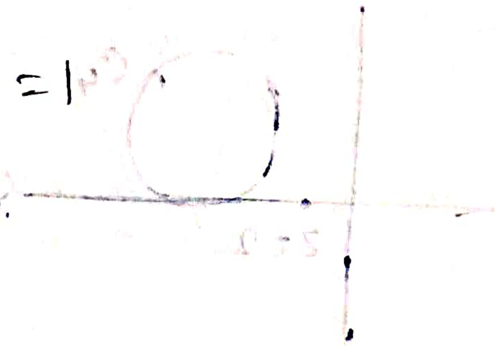
$$= 2\pi i \int_0^1 dt = 2\pi i \neq 0$$

1. $I_1 \neq 0$.
2. I_1 is independent of R.
3. Any closed path enclosing origin will have $I_1 = 2\pi i$.
4. Something special at origin $z = 0$?

Check :- Do this integration $\oint \frac{1}{z^2} dz$ circle of Radius R.

Ex-2 :-

$$I_2 = \oint z^2 dz \,; \quad C: |z| = 1$$



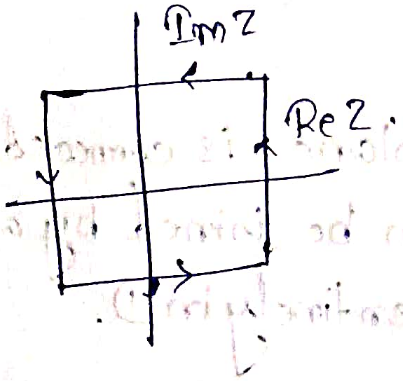
$$= \int_0^{2\pi} e^{2i\theta} \, i e^{i\theta} \, d\theta$$

$$= i \int_0^{2\pi} e^{3i\theta} \, d\theta .$$

$$= \frac{i}{3i} \left[e^{3i\theta} \right]_0^{2\pi}$$

$$= \frac{1}{3} \left[e^{6i\pi} - 1 \right] = 0 .$$

- $f(z) = z^2$ is entire function.
- Any closed curve, $I_2 = 0$

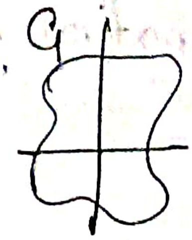


Cauchy's Theorem :-

If $f(z)$ is analytic in a simply connected domain D and C be a closed contour in D.

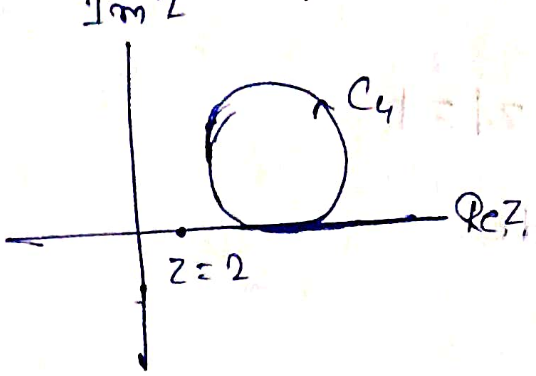
$$\oint f(z) dz = 0$$

3. $I_3 = \oint_{C_1} e^{z} dz$



$= 0.$

4. $\oint_{C_4} \frac{e^{z^2}}{z-2} dz$



$\Rightarrow$ At $z = 2$ function is not analytic (as it is not defined at $z=2$).

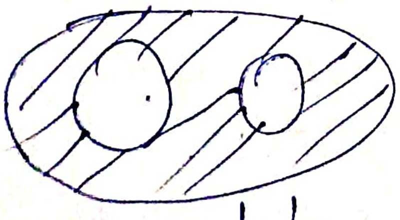
$$I_4 = \oint_{C_4} \frac{e^{z^2}}{z-2} dz = 0$$

(As the $z = 2$ is outside the contour C_4. In the contour C_4, $f(z) = \frac{e^{z^2}}{z-2}$ is analytic).

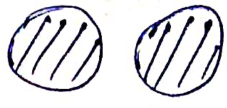
Def^n (Connected) :-

A set D of points in $\mathbb{C}$ plane is connected if any two points in D can be joined by a continuous curve contained entirely in D.

$\rightarrow$ connected



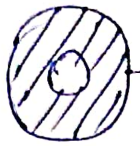
connected.



$\hookrightarrow$ not connected.

Defⁿ (Simply Connected) :-

A connected domain is simply connected if every continuous closed curve in D can be continuously deformed to a single point without going outside D.



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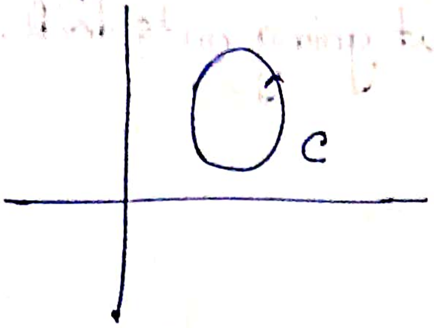
Cauchy's Theorem

If $f(z)$ is analytic inside a closed contour C and on C



$$\oint_C f(z)\,dz = 0.$$

Proof :- $f(z)$ also continuous.

Green's Theorem :-

$$\boxed{\oint (p\,dx + q\,dy) = \iint_R \left(\frac{\partial q}{\partial x} - \frac{\partial p}{\partial y}\right) dx\,dy.}$$

$$\oint f(z)\,dz = \int (u\,dx - v\,dy) + i\int (v\,dx + u\,dy).$$

$$= \iint \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dx\,dy + i\iint \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dx\,dy$$

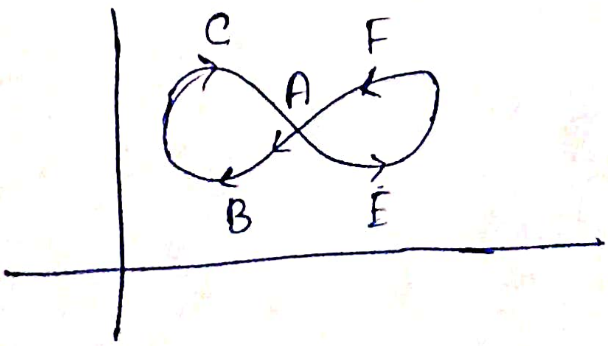
$= 0$ [By CR conditions].

$\oint z^2 dz = 0$. (Using Cauchy's Theorem).

$\oint z^2 dz \rightarrow$ Check explicitely

$C: |z| = 1$

$C = \square$, $\oint z^2 dz = 0$.



$C_1 = AEFA \rightarrow$ anticlockwise

$C_2 = ABCA$

$\hookrightarrow$ ~~anti~~clockwise.

$$\oint_{C_1} f(z)\,dz = 0$$

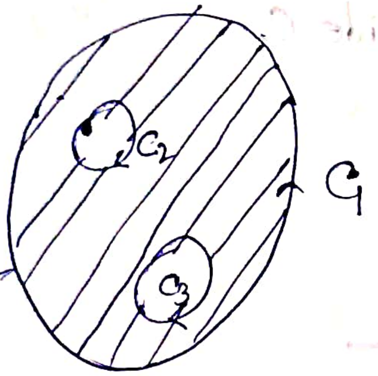
$$\oint_{-C_2} f(z)\,dz = 0 \Rightarrow -\oint_{C_2} f(z)\,dz = 0$$

$$\oint_{C} f(z)\,dz = \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz$$

$$= 0 + 0 = 0$$

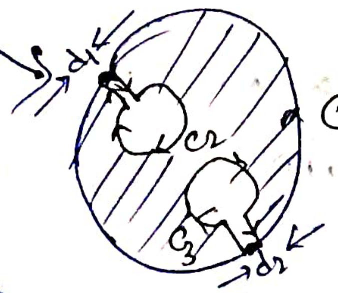
$\oint \frac{1}{z}\,dz$  contour $\neq 0$ (not analytic in 0).

 $\oint \frac{1}{z}\,dz = 0$ (analytic inside the contour)



Domain of analyticity of function is shaded region ▨.

$\int_{C_1} f(z)\,dz \longrightarrow$ can not use Cauchy's Theorem (C_2 and $C_3 \to$ function is non analytic)



$C_1 \to C_1'$.

$$\oint_{C_1'} f(z)\,dz = 0$$

$= \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz + \oint_{C_3} f(z)\,dz +$ extra pieces (d_1 & d_2)

$\Downarrow$
0 as d_1 & $d_2 \to 0$.

$$\oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz + \oint_{C_3} f(z)\,dz = 0 .$$

It happens only when
↳ Inside contoum is running opposite dimection to the whole outside contoum.

$$\Rightarrow \sum_{i=C_i}^{n} \oint f(z) = 0 .$$

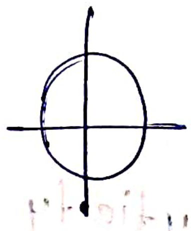
Cauchy's Integral Formula

Let $f(z)$ be analytic in D & C be a closed contour in D

$$\oint_C \frac{f(z)\,dz}{z-z_0} = \begin{cases} 0 , & z_0 \text{ outside } C . \\ 2\pi i f(z_0) , & z_0 \text{ inside } C \end{cases}$$

↳ verification :-

1. $\oint \frac{1}{z}\,dz$



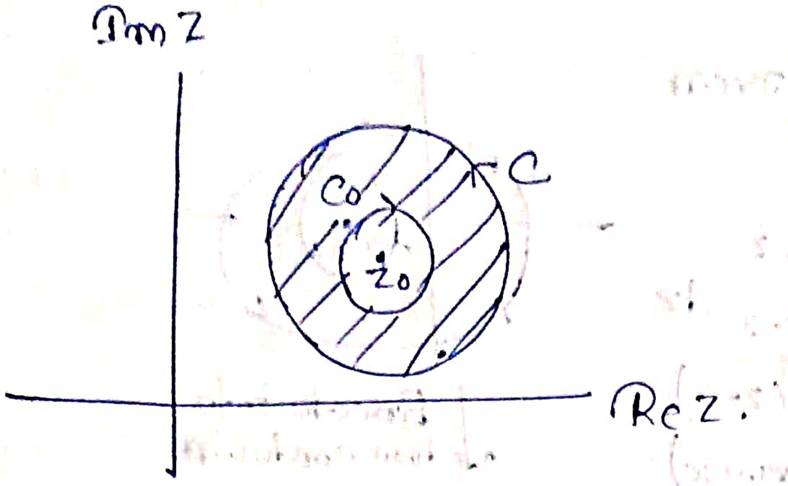
$= \oint \frac{f(z)}{z-0}\,dz$, with $f(z) = 1$

$= 2\pi i f(z=0) = 2\pi i .$

2. $\oint \frac{1}{z^2}\,dz$

$= \oint \frac{f(z)}{z-0}\,dz$, with $f(z) = \frac{1}{z}$

↳ We cannot use this formula as $f(z)$ is not analytic in its domain



$$\oint_C \frac{f(z)}{z-z_0}\,dz + \oint_{C_0} \frac{f(z)}{z-z_0}\,dz = 0.$$

$$\Rightarrow \oint_C \frac{f(z)}{z-z_0}\,dz = -\oint_{C_0} \frac{f(z)}{z-z_0}\,dz = -\int \frac{f(\varepsilon,\theta)}{\varepsilon} e^{-i\theta} \varepsilon e^{i\theta}\, i\, d\theta$$

C_0 : circle of radius ε. $\quad = -\int f(\varepsilon,\theta)\, i\, d\theta.$

$z - z_0 = \varepsilon e^{i\theta}.$

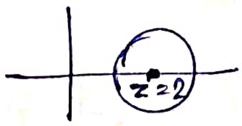
Now, if $f(z)$ is continuous & take $\varepsilon \to 0$;

$f(z) = f(z_0)$

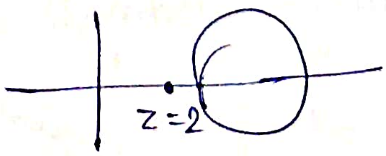
$$\Rightarrow \oint_C \frac{f(z)}{z-z_0}\,dz = -f(z_0)\, i \int d\theta = -f(z_0)\, i\, (-2\pi).$$

$$= 2\pi i f(z_0).$$

Example :-

1. $\oint \frac{e^{z^2}}{z-2}\,dz.$ over 

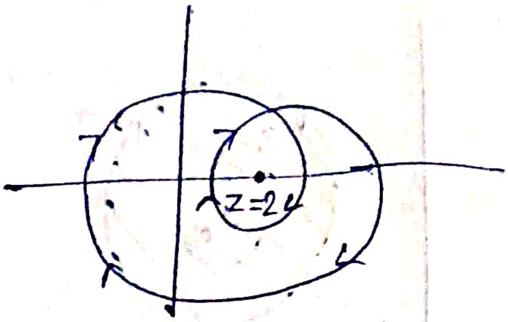
$= 2\pi i f(2) = 2\pi i e^4.$

~~But~~ But if contour is 

$I = 0.$

2.

2. $\oint_C \frac{e^z}{z-2}\, dz$ over



$z_0 = 2$

$= \oint_{C_1} \frac{e^z}{z-2}\, dz + \oint_{C_2} \frac{e^z}{z-2}\, dz$

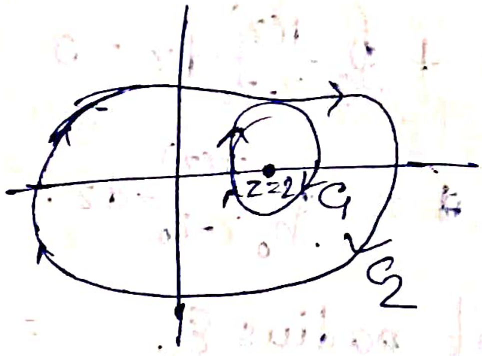
$= -2\pi i f(z=2) - 2\pi i f(z=2)$

($\because$ both are clockwise)

$= -4\pi i f(z=2)$

$= -4\pi i e^2$

Break into two contours.



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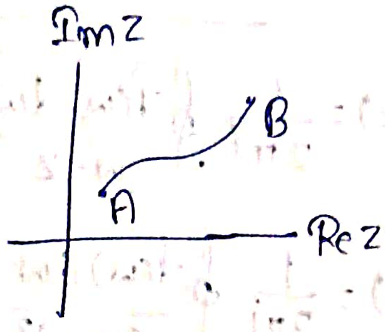
- $\int_a^b f(z)\,dz$

$= \int (\quad)\,dx + \int (\quad)\,dy$



- **Cauchy's Theorem :-**

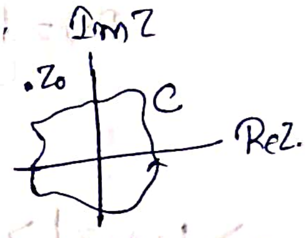
$\oint_C f(z)\,dz = 0$

If $f(z)$ is analytic inside C.

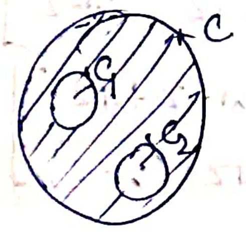
- **Cauchy's Integral Formula :-**

when $f(z)$ is analytic in domain D,

$$\oint_C \frac{f(z)}{z-z_0}\,dz = \begin{cases} 0 & \text{if } z_0 \text{ outside } C. \\ 2\pi i f(z_0) & \text{if } z_0 \text{ inside } C. \end{cases}$$



$g(z) = \frac{f(z)}{z-z_0}$, non analytic at $z = z_0$.



$\oint_C f(z)\,dz + \oint_{C_1} f(z)\,dz + \oint_{C_2} f(z)\,dz = 0$.

Derivative of an Analytic Function :-

$\left|\oint_C f(z)\,dz\right| \le ML \longrightarrow$ Darboux Inequality.

Given $|f(z)| \le M$, L is the length of the contour

Proof :-

$|I| = \left|\sum_{j=1}^{n} f(\zeta_j)\,\Delta z_j\right| \le \sum_{j=1}^{n} |f(\zeta_j)\,\Delta z_j|$

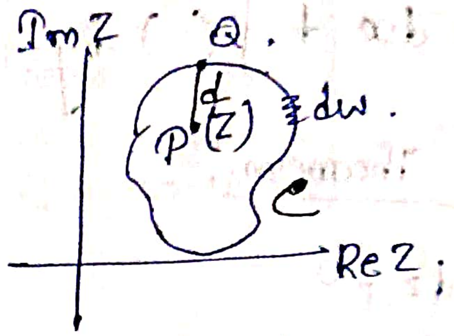
$= \sum_{j=1}^{n} |f(\zeta_j)||\Delta z_j| \le M \sum |\Delta z_j|$

$= ML$. (Proved)

$|\sum z_n| \le \sum |z_n|$ and $|z_1+z_2| \le |z_1| + |z_2|$.

Derivative of an Analytic Function

$f(z) = \frac{1}{2\pi i} \oint \frac{f(w)\,dw}{w-z}$, where z is inside C.



$f'(z) = \frac{1}{2\pi i} \oint \frac{f(w)\,dw}{(w-z)^2}$.

↳ We have to check wheather it is correct or not.

- z is a point interior of C.
- Q ~~is~~ is another point on C closest to P.

$PQ = d > 0$ → smallest distance of z from the contour.

$\Rightarrow |w-z| \geq d$ for all w on C.

$$\frac{f(z+\Delta z) - f(z)}{\Delta z} = \frac{1}{2\pi i} \frac{1}{\Delta z} \oint dw\, f(w) \left[\frac{1}{w-z-\Delta z} - \frac{1}{w-z} \right]$$

$$= \frac{1}{2\pi i \Delta z} \oint dw\, f(w) \left[\frac{w-z-w+z+\Delta z}{(w-z-\Delta z)(w-z)} \right].$$

$$= \frac{1}{2\pi i} \oint \frac{dw\, f(w)}{(w-z-\Delta z)(w-z)}.$$

$$\Delta I = \frac{1}{2\pi i} \oint_C dw\, f(w) \left[\frac{1}{(w-z-\Delta z)(w-z)} - \frac{1}{(w-z)^2} \right].$$

If we can show $\Delta I \to 0$ as $\Delta z \to 0$ then it ~~is~~ ~~obvious~~ ~~that~~ follows that —

$$f'(z) = \frac{1}{2\pi i} \oint \frac{f(w)\,dw}{(w-z)^2}$$

$$\Delta I = \frac{1}{2\pi i} \oint dw\, f(w) \, \frac{(w-z)-(w-z-\Delta z)}{(w-z-\Delta z)(w-z)^2}$$

$$= \frac{\Delta z}{2\pi i} \oint \frac{dw\, f(w)}{(w-z-\Delta z)\cdot(w-z)^2}$$

So, it is clear that $\Delta I \to 0$ as $\Delta z \to 0$ if

$\oint \frac{dw\, f(w)}{(w-z-\Delta z)(w-z)^2}$ is finite.

$|w-z| \geq d$.

$|w-z-\Delta z| \geq \big| |w-z| - |\Delta z| \big| \geq |w-z| - |\Delta z|$.

$$\left| \oint \frac{dw\, f(w)}{(w-z-\Delta z)(w-z)^2} \right| \leq \frac{ML}{d^2(d-\Delta z)}.$$

$$\boxed{f'(z) = \frac{1}{2\pi i} \oint \frac{dw\, f(w)}{(w-z)^2}.}$$

$$f^n(z) = \frac{n!}{2\pi i} \oint \frac{f(w)\, dw}{(w-z)^{n+1}}$$ if z inside C.

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$$f(z) = \frac{1}{2\pi i} \oint_C \frac{f(w)}{(w-z)}\, dw.$$

where $w = z$ inside C.

$$f^n(z) = \frac{n!}{2\pi i} \oint_C \frac{f(w)\, dw}{(w-z)^{n+1}}$$

where z inside C.

(*) Differentiability of a real function $f(x)$ does not guarantee that it is going to be infinitely differentiable.

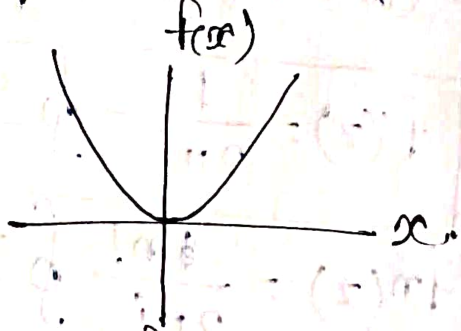
Ex 1:- $f(x) = x^{\frac{4}{3}}$ $(x \neq 0)$; 0 for $x = 0$.

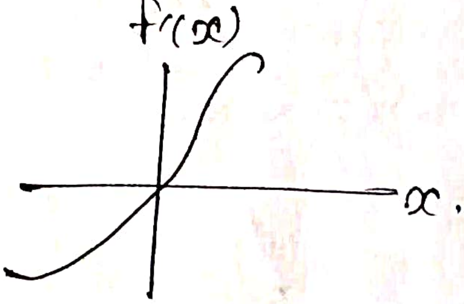
$$f'(0) = \lim_{h \to 0} \frac{h^{\frac{4}{3}} - 0}{h} = 0.$$

$$f'(x) = \frac{4}{3} x^{-\frac{2}{3}}$$

$$\lim_{x \to 0} f''(0) \to \infty.$$

↳ 2^{nd} derivative does not exist.





Ex 2:- $f(x) = \begin{cases} x^2 \sin\frac{1}{x}, & x \neq 0 \\ 0, & x = 0. \end{cases}$

$\Rightarrow$ 2^{nd} derivative does not exist (Check!).

(*) Complex Analytic function $f(z)$ is infinitely differentiable.

Gauss's Mean Value Theorem

- C be a circle $|z-z_0| = R$ in the domain of analyticity of $f(z)$. Then $f(z_0) = \frac{1}{2\pi}\int_0^{2\pi} f(z_0 + Re^{i\theta})\, d\theta$.

Proof :- Use Cauchy's theorem and then use polar coordinate. (Homework)

Theorem (Cauchy Liouville Theorem) :-

A bounded ($|f(z)| \leq M$) entire function must be constant.

Ex 1 :- $f(z) = 7 \rightarrow$ bounded
Entire.
$\hookrightarrow$ Constant.

Ex 2 :- $f(z) = e^{iz} \rightarrow$ entire
$\hookrightarrow$ Not constant as not bounded.

$$|f(z)| = |e^{-y+ix}| = e^{-y}$$

$\hookrightarrow$ diverge at $y \rightarrow -\infty$.

(*) Non constant Analytic functions either have singularities or unbounded.

Proof :-

$|f(z)| \leq M$.

$\Rightarrow \left|\oint_C f(z)\,dz\right| \leq ML \longrightarrow$ Darboux Inequality.

$\Rightarrow \frac{1}{2\pi i}\oint_C \frac{f(z')}{(z'-z)^2}\,dz' = f'(z) \longrightarrow$ Take C be a circle $|z'-z_0| = R$

z lies inside C.

$|f'(z)| \leq \frac{M}{2\pi R^2}\, 2\pi R = \frac{M}{R}$.

Now f(z) is entire.

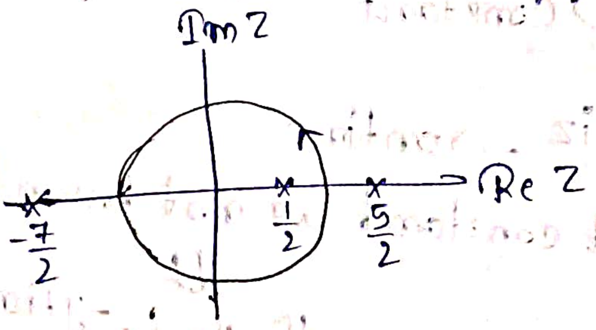
True for any value of R.

$\Rightarrow f'(z) = 0$

$\Rightarrow$ f(z) has to be constant. (Proved).

Ex 1 :-
$$\oint_C \frac{z^2\,dz}{\left(z-\frac{1}{2}\right)\left(z+\frac{7}{2}\right)\left(z-\frac{5}{2}\right)}.$$

$\Rightarrow$



$\oint \frac{f(z)\,dz}{z-z_0} = \cancel{f(z_0)}\ 2\pi i f(z_0)$ if z_0 inside C

$C: |z| = 1$.

$= \oint \frac{f(z)}{\left(z-\frac{1}{2}\right)}\,dz$ where $f(z) = \frac{z^2}{\left(z+\frac{7}{2}\right)\left(z-\frac{5}{2}\right)} \rightarrow$ Not entire function

$\hookrightarrow$ non analytic points are outside of the contour.

$= 2\pi i\, f\left(z = \frac{1}{2}\right)$.

• $C_2 : |z| = \frac{1}{5} \Rightarrow I = 0$

• $C_3 : |z| = 3 \Rightarrow$

$$\oint \frac{z^2}{\left(z+\frac{7}{2}\right)} \frac{1}{\left(z-\frac{1}{2}\right)\left(z-\frac{5}{2}\right)}$$

$$= \oint \frac{z^2}{\left(z+\frac{7}{2}\right)} \left[\frac{1}{\left(z-\frac{5}{2}\right)} - \frac{1}{\left(z-\frac{1}{2}\right)}\right]$$

$$= \frac{1}{2}\, 2\pi i \left[f\left(z=\tfrac{5}{2}\right) + f\left(z=\tfrac{1}{2}\right)\right]$$

$$= \frac{1}{2} \times 2\pi i \left[\frac{\left(\frac{5}{2}\right)^2}{6} + \frac{\frac{1}{4}}{4}\right]$$

$$= \pi i \left[\frac{25}{24} + \frac{1}{16}\right]$$

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Cauchy's Theorem :-

$\oint_C f(z)\,dz = 0$ if $f(z)$ analytic inside and on C.

Morea's Theorem :-

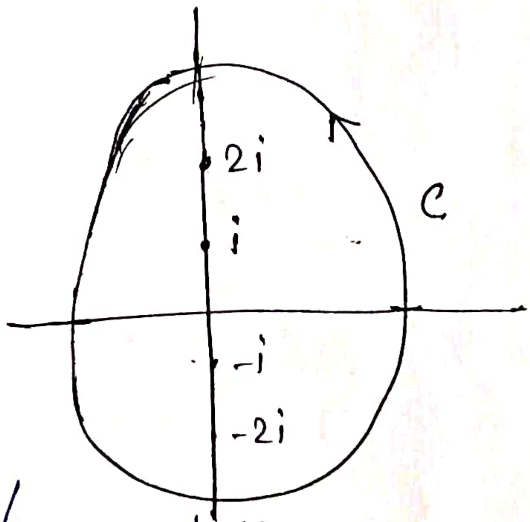
If $\oint_C f(z)\,dz = 0$ for all closed contours in D.
then $f(z)$ is analytic.

• For multiply connected domain :-

$$\frac{1}{2\pi i}\left[\oint_C [\;] + \oint_{C_2}[\;] + \oint_{C_3}\left[\frac{f(z')}{z'-z_0}dz'\right]\right] = \begin{cases} f(z_0) & \text{if } z_0 \text{ inside } D. \\ 0 & \text{if } z_0 \text{ outside } D. \end{cases}$$

$$[\;] \longrightarrow \left[\frac{f(z')}{z'-z_0}dz'\right].$$

Example :-



1) $I = \oint_C \frac{z}{z^2+4}\,dz$ along
$C : |z| = 3$.

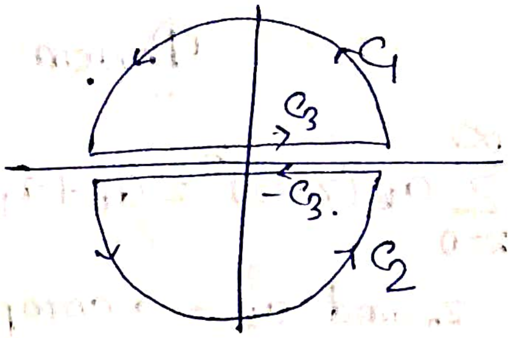
$\Rightarrow z = \pm 2i$, $f(z)$ singular/non analytic

$z_0^{(1)} = 2i$, $z_0^{(2)} = -2i$.

$$f(z) = \frac{z}{(z-2i)(z+2i)}$$

$$f_1 = \frac{z}{z+2i}, \quad f_2 = \frac{z}{z-2i}$$

$$f(z) = \frac{f_1(z)}{z-2i} = \frac{f_2(z)}{z+2i}$$



$$\oint_C \frac{z}{z^2+4} dz = \int_{C_1+C_3-C_3+C_2} \left(\frac{z}{z^2+4}\right) dz.$$

$$= \oint_{C_1+C_3} \frac{f_1(z)}{z-2i} dz + \oint_{C_2-C_3} \frac{f_2(z)}{z+2i} dz$$

$$= f_1(z=2i) + f_2(z=-2i).$$

$$= 2\pi i \left[f_1(z=2i) + f_2(z=-2i)\right].$$

$$= 2\pi i \left[\frac{1}{2} + \frac{1}{2}\right] = 2\pi i.$$

- Domain $D = \mathbb{C} \setminus \{0\}$.

$f(z) = \frac{1}{z}$.

$$\oint \frac{\frac{1}{z}}{z-2} = 2\pi i f(z=2) = 2\pi i \times \frac{1}{2} = \pi i.$$

$C: |z| = 3$.

▨ If $D \equiv \mathbb{C}$, we can not use Cauchy's Integral formula.

2) $\oint_C \frac{z^3}{(z+1)^3} dz$ where $C: |z| = 3$.

$$= 2\pi i f''(z=-1)$$

$$= -2\pi i \times 3 \times \frac{2}{2} = -6\pi i.$$

$2\pi i f(z) = \oint \frac{f(z')}{z'-z} dz'$ if z inside C

$2\pi i f''(z) = 2\oint \frac{f(z')}{(z'-z)^3} dz'.$

Power Series

$$\sum_{k=0}^{\infty} a_k (z-z_0)^k = a_0 + a_1 (z-z_0) + a_2 (z-z_0)^2 + \cdots$$

z_0 and $a_k \longrightarrow$ complex number.

Ex :-

$1+z+z^2+ \cdots$ (Geometric Series).

$z_0 = 0$, $a_k = 1 \ \forall k$.

converges to $|z|<1$.

Theorem :-

A power series $\sum_{k=0}^{\infty} a_k (z-z_0)^k$ with radius of convergence $R>0$ represents an analytic function $f(z)$ in the interior of R.

- Let $z=z_0$ a point in the domain of analyticity of a function $f(z)$, then there exist a Unique power series expansion –

$$f(z) = \sum_{n=0}^{\infty} a_n (z-z_0)^n \longrightarrow \text{valid in the largest disk around } z_0.$$

$$a_n = \frac{1}{n!} \left. \frac{d^n f}{dz^n} \right|_{z=z_0}$$

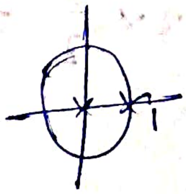
Taylor Series.

Example :-

1) Geometric Series :-

$$f(z) = \frac{1}{1-z} = 1+z+z^2+z^3+ \cdots$$

$z_0 = 0$



2) $f(z) = \dfrac{1}{(z-2)^2}$

$= \dfrac{1}{4}\left(1 - \dfrac{z}{2}\right)^{-2}$

$= \dfrac{1}{4}\left[1 + 2\cdot\dfrac{z}{2} + 2.3\cdot\left(\dfrac{z}{2}\right)^2 + \cdots\right]$

$z_0 = 0.$

↳ Radius of convergence $= 2$.

$a_n = \dfrac{n+1}{4}\left(\dfrac{1}{2}\right)^n.$

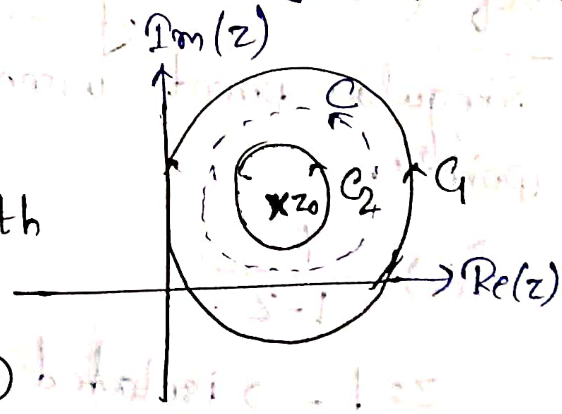
3) $f(z) = e^z.$

$= 1 + z + \dfrac{z^2}{2} + \cdots$

$z_0 = 0$, $a_n = \dfrac{1}{n!}$

$R \to \infty$. ⟶ no point of non ~~analit~~ analyticity

Theorem :-



· Let C_1 and C_2 concentric circles with centre $z = z_0$. $f(z)$ analytic between C_1 and C_2. and on C_1 and C_2 $\} \Rightarrow D$

↳ annular region.

Then in D, any point ~~f(z)~~ z, $f(z)$ is given by a convergent power series

$$f(z) = \sum_{n=-\infty}^{\infty} d_n (z-z_0)^n . \left(= d_0 + d_1(z-z_0) + d_{-1}(z-z_0)^{-1} + d_2(z-z_0)^2 + d_{-2}(z-z_0)^{-2} + \cdots\right).$$

where $d_n = \dfrac{1}{2\pi i}\oint_C \dfrac{f(z')}{(z'-z_0)^{n+1}}\, dz'$, where C any contour in D.

⟶ Laurent Expansion.

Example :-

1) $f(z) = e^{\frac{1}{z}}$.

$$= \sum_{n=0}^{\infty} \frac{1}{n!} \frac{1}{z^n}$$

$$= 1 + \frac{1}{z} + \frac{1}{2z^2} + \frac{1}{6z^3} + \cdots$$

$d_n = 0$ for $n > 0$ [We define s.t. inverse term $= d_{-n}$].

$d_{-n} \neq 0 \ \forall n$ | $\to z = 0$ ↳ Essential Singularity.

- $f(z_0)$ analytic $\Rightarrow z = z_0$ regular
- $f(z_0)$ non-analytic $\Rightarrow z = z_0$ singular.

Def^n (Isolated Singular Points) :-

Singular point surrounded by regular points.

$$f(z) = \frac{1}{1-z}.$$

$z = 1 \longrightarrow$ isolated singularity.

- If d_{-N} $(N > 0)$ and $d_{-n} = 0$ for $n < -N$, $f(z)$ has Nth order pole. Highest negative power be N.

Ex :- $f(z) = \dfrac{1+2z}{z^2} = \dfrac{2}{z} + \dfrac{1}{z^2}$.

$$= \frac{2}{(z-0)} + \frac{1}{(z-0)^2} + \cdots$$

$1 = d_{-2}$, $2 = d_{-1}$ Second order pole (highest power z^2 in the expansion)

• If $N=1 \Rightarrow$ Simple Pole, $d_{-1} \neq 0$, ~~$d_{-2} \neq 0$~~ $d_{-2} = 0 \ldots$

$N=2 \Rightarrow$ double pole,

$d_{-2} \neq 0, \; d_{-3} = d_{-4} = d_{-5} = \cdots = 0.$

Example :-

1) $f(z) = \frac{1}{z} \frac{1}{(1-z)^2}$ → expand it around $z=0$.

$$= \frac{1}{z}\left(1+2z+3z^2+4z^3+\cdots\right).$$

$$= \frac{1}{z} + 2 + 3z + 4z^2 + \cdots$$

$$\left.\begin{aligned} d_{-1} &= 1 \\ d_{-2} &= 0 \\ d_1 &= 3 \\ d_0 &= 2 \\ d_2 &= 4 \end{aligned}\right\} \longrightarrow \text{Order of pole} = 1.$$

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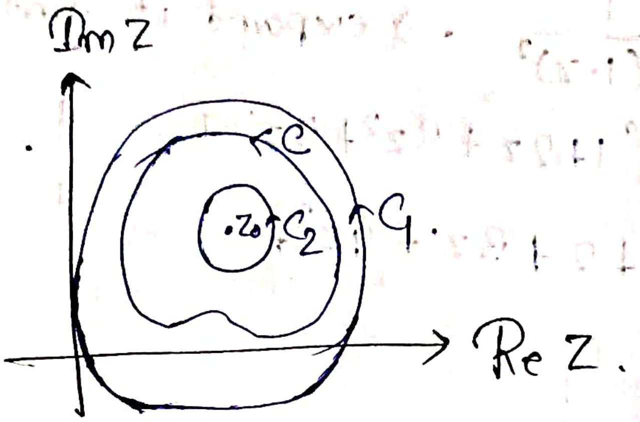
Power Series → 1) Taylor Series
→ 2) Laurent Series.



$$f(z) = \sum_{n=-\infty}^{\infty} d_n (z-z_0)^n$$

$$d_n = \frac{1}{2\pi i} \oint_C \frac{f(z')}{(z'-z_0)^{n+1}} dz'$$

$$= \sum_{n=0}^{\infty} a_n (z-z_0)^n + \sum_{n=0}^{\infty} b_n (z-z_0)^{-n}.$$

$$\rightarrow a_n = \frac{1}{2\pi i} \int_C \frac{f(z)\, dz}{(z-z_0)^{n+1}}$$

$$b_n = \frac{1}{2\pi i} \int_C \frac{f(z)\, dz}{(z-z_0)^{-n+1}}$$

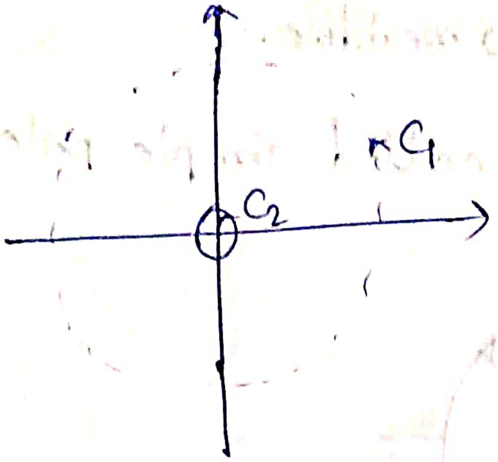
Example

1. $f(z) = \frac{1+2z}{z^2}$

Do Laurent Series Expansion around $z_0 = 0$.

$\Rightarrow f(z) = \frac{1}{z^2} + \frac{2}{z}.$

$z_0 = 0$ singular point.



$C_2 : R = \varepsilon.$
$C_1 : R = \infty.$

$|z| > 0$

Regular part is the part coming from Taylor series.

$$\left.\begin{array}{l} d_{-2} = 1 \\ d_{-1} = 2 \\ d_{-3} = 0 \end{array}\right\} \longrightarrow 2^{nd} \text{ order pole.}$$

$d_{-1} \equiv$ Residue of $f(z)$.

2. $f(z) = \dfrac{1}{z^2}$

$\Rightarrow$ function is not analytic at $z=0$.

$\Rightarrow d_{-1} = 0 \Rightarrow$ Residue $= 0$.

3. $f(z) = \dfrac{1}{z(z-1)}$

$\Rightarrow$ Singular at $z=0$ and $z=1$.

$\Rightarrow$ Series expansion around $z_0 = 0$.

A. Do Laurent Series expansion around $z_0 = 0$.

$$\Rightarrow f(z) = \frac{1}{z-1} - \frac{1}{z}$$

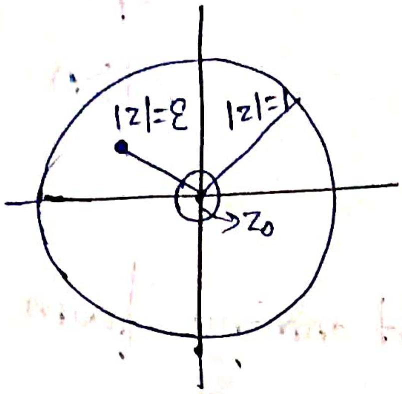
$$= -\frac{1}{z} - \frac{1}{(1-z)}$$

$$= -\frac{1}{z} - (1-z)^{-1} = -\frac{1}{z} - [1+z+z^2+ \dots] \quad (I_1, |z|<1)$$

$$= -\frac{1}{z} - 1 - z - z^2 - \dots$$

$d_{-1} = -1 \Rightarrow$ residue.

1^{st} order pole are called simple pole.



Region of validity - $0 < |z| < 1$.

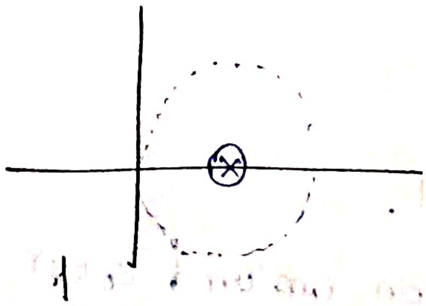
B. Do expansion around $z_0 = 1$.

$$f(z) = -\frac{1}{z} + \frac{1}{z-1}$$

$$= \frac{-1}{1-(1-z)} + \frac{1}{z-1}$$

$$= -1\left[1-(1-z)\right]^{-1} - \frac{1}{1-z}$$

$$= -\sum_{n=0}^{\infty}(1-z)^n - \frac{1}{1-z} \qquad |1-z| < 1.$$



$d_{-1} = 1$.

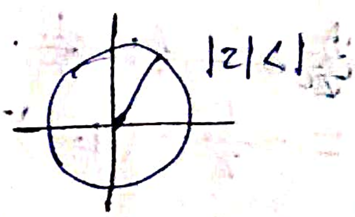
3. $f(z) = \frac{1}{z-1}$

A. Do expansion around $z_0 = 0$.

$$= -\sum_{n=0}^{\infty} z^n \qquad |z| < 1.$$

↓

Taylor Series.



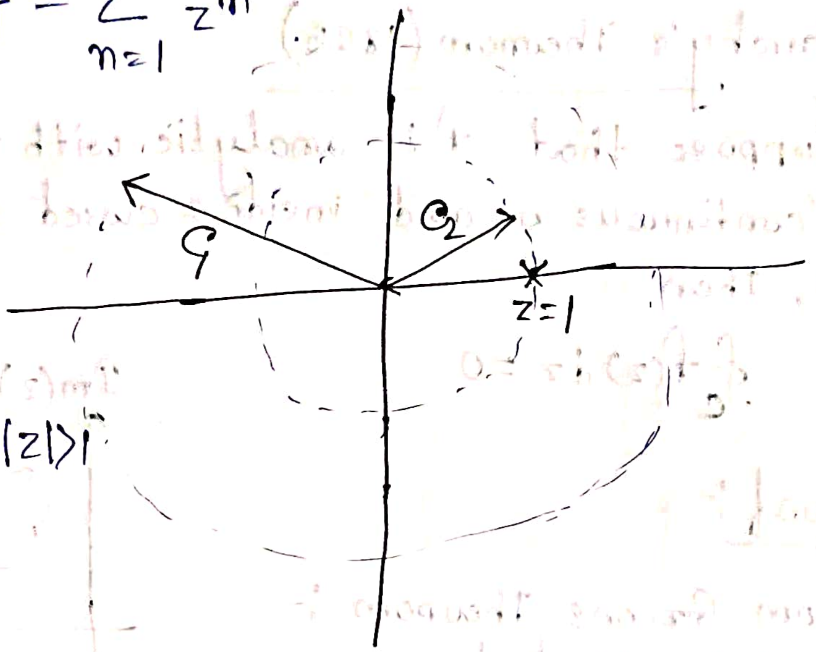
B. Do an expansion valid for $|z|>1$.

$$f(z)=\frac{1}{z\left(1-\frac{1}{z}\right)}.$$

$$=\frac{1}{z}\left(1-\frac{1}{z}\right)^{-1} \longrightarrow \left|\frac{1}{z}\right|<1 \Rightarrow |z|>1.$$

$$=\sum_{n=0}^{\infty}\frac{1}{z^{n+1}}=\sum_{n=1}^{\infty}\frac{1}{z^{n}}$$

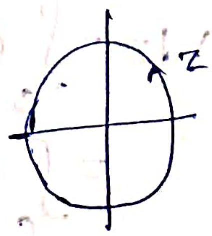
$z_0=0 \rightarrow$ not analytic.



Residue Theorem :-

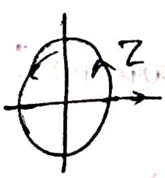
Let $f(z)$ be analytic within and on a closed contour C except finite number of singular points $z_1, z_2, \ldots, z_n$ inside C

$$\oint_C f(z)\,dz = 2\pi i \sum_{j=1}^{m} \operatorname{Res} f(z)\Big|_{z=z_j}$$



Example :-

1) $\oint_C \frac{1}{z^2}\,dz$, $C: |z|=1$



$\Rightarrow f(z)=\frac{1}{z^2} \Rightarrow d_{-1}=0$

$\hookrightarrow$ Residue $=0$

$\Rightarrow \oint_C \frac{1}{z^2}\,dz = 0$.

$\oint_C z^2\,dz = 0$

$C: |z|=1$

2. $\oint_C \frac{1}{z} dz \quad C: |z| = 1$

$= 2\pi i \times 1 \quad [\text{Residue} = 1]$.

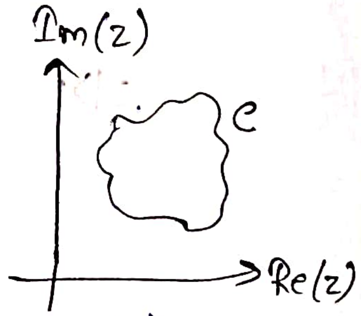
$= 2\pi i$

21/08/2025

Cauchy's Theorem (1825)

Suppose that f is analytic with the derivative f' continuous on and inside a closed simple contour C, Then

$$\oint_C f(z)\, dz = 0$$



$\oint_C f(z)\, dz = 0$

Proof :-

From Greens Theorem :-

it states that for continuously differentiable functions $P(x, y)$ and $Q(x, y)$

$$\int_C P\,dx + Q\,dy = \iint_R \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dA$$

$f = u + iv$.

$$\int_C f(z)\, dz = \int_C (u + iv)(dx + i\,dy)$$

$$= \int_C (u\,dx - v\,dy) + i(u\,dy + v\,dx)$$

By applying Green's Theorem :

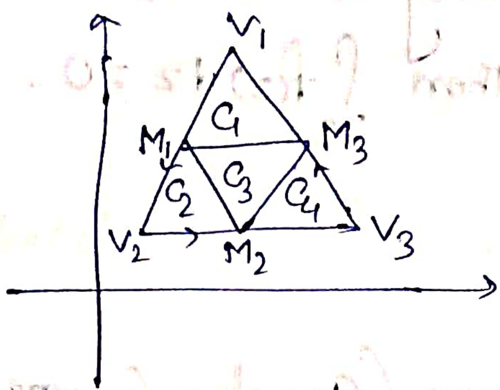
$$\int_C f(z)dz = \iint_R \left(-\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right) dA + i \iint_R \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right) dA$$

$= 0$.

Lemma 1 :-

If Δ is a triangular contour entirely within D, then $\int_{\Delta} f(z)\,dz = 0$.

Proof :-



$$\int_{\Delta} f(z)\,dz = \int_{C_1} f(z)\,dz + \int_{C_2} f(z)\,dz + \int_{C_3} f(z)\,dz + \int_{C_4} f(z)\,dz .$$

$$\left|\int_{\Delta} f(z)\,dz\right| \leq \left|\int_{C_1} f(z)\,dz\right| + \left|\int_{C_2} f(z)\,dz\right| + \left|\int_{C_3} f(z)\,dz\right| + \left|\int_{C_4} f(z)\,dz\right|$$

$$\leq 4\left|\int_{\Delta_1} f(z)\,dz\right| .$$

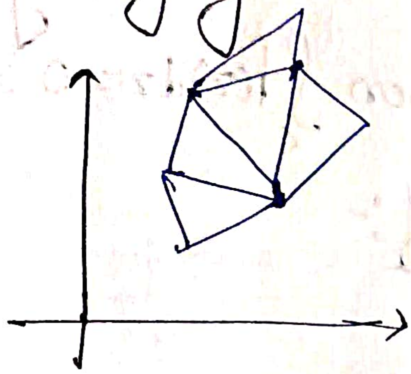
$$\left|\int_{\Delta_1} f(z)\,dz\right| = \max_{\substack{C_i \\ i\in\{1,2,3,4\}}} \left|\int_{C_i} f(z)\,dz\right| .$$

Repeating the same process -

Lemma 2 :-

If C is a closed polygonal contour lying entirely within D

Then $\int_C f(z)\,dz = 0$.



Theorem (Cauchy Goursat Theorem) :-

If a function is analytic at all points interior to and on a simple closed contour C, then

$$\int_C f(z)\,dz = 0$$

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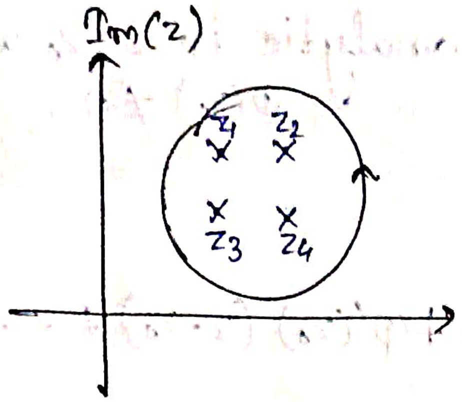
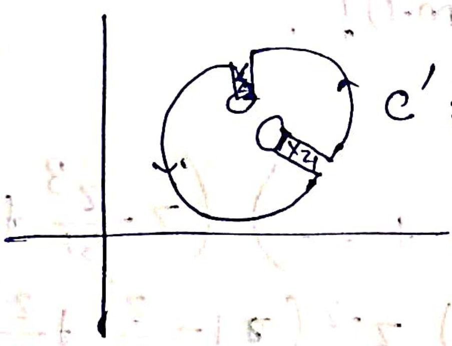
Residue Theorem



$$\oint_C f(z)\,dz = 2\pi i \sum \text{Res } f(z)\Big|_{z=z_j}$$

Proof :- A different contour —



$C' \equiv$ new contour.

$$\oint_{C'} f(z)\,dz = 0$$

$$\Rightarrow \oint_C f(z)\,dz + \sum_{j=1}^{n} \oint_{C_j} f(z)\,dz + \underbrace{\int_{\text{lines}} f(z)\,dz}_{0} = 0.$$

$$\Rightarrow \oint_C f(z)\,dz - 2\pi i \sum_{j=1}^{n} \text{Res } f(z)\Big|_{z=z_j} = 0$$

$d_n = \frac{1}{2\pi i}\oint \frac{f(z)\,dz}{(z-z_0)^{n+1}}$

d_{-1} :- residue

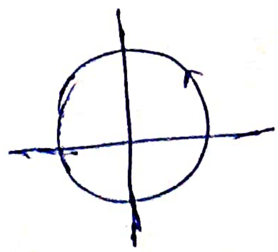
$$\Rightarrow \oint_C f(z)\,dz = 2\pi i \sum_j \text{Res } f(z)\Big|_{z=z_j}$$

Example :-

1) $\oint \frac{1}{z^2}\,dz = 0$

2) $\oint \frac{1}{z}\,dz = 2\pi i \times 1 = 2\pi i$

$C: |z| = 2$



I. $f(z) = \dfrac{\varphi(z)}{(z-z_0)^m}$ $\quad z \neq z_0$

where $\varphi(z)$ analytic in $z = z_0$ and $\varphi(z_0) \neq 0$.

$$f(z) = \frac{\cos z}{\sin^2 z}.$$

$$f(z) = \frac{1}{(z-z_0)^m}\left[\varphi(z_0) + \varphi'(z_0)(z-z_0) + \cdots\right].$$

$$= \frac{\varphi(z_0)}{(z-z_0)^m} + \cdots + \frac{\varphi^{m-1}(z_0)}{(m-1)!\,(z-z_0)}$$

Residue :- $\dfrac{\varphi^{m-1}(z_0)}{(m-1)!}$.

Ex :-

1) $f(z) = \left(1 - \dfrac{z^2}{2!} + \dfrac{z^4}{4!} + \cdots\right)\left(z - \dfrac{z^3}{3!} + \dfrac{z^5}{5!} - \cdots\right)^{-2}$

$$= (1 - \cdots)\, z^{-2}\left(1 - \frac{z^2}{3!} + \frac{z^4}{5!} - \cdots\right)$$

$$= \frac{1}{z^2}\left(1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \cdots\right)\left(1 + \frac{2z^2}{3!} + \cdots\right).$$

$$= \frac{1}{z^2}\left(1 + \frac{2}{3}z^2 + \cdots\right)$$

$$= \frac{1}{z^2} + \frac{2}{3} + \cdots$$

$\longrightarrow$ No $\left(\frac{1}{z}\right)$ term

Residue $= 0$.

$$* \operatorname{Res} f(z)\Big|_{z=z_0} = \frac{1}{(m-1)!}\,\frac{d^{m-1}}{dz^{m-1}}\Big[(z-z_0)^m f(z)\Big]\Big|_{z=z_0}.$$

$$\frac{d^{m-1}}{dz^{m-1}}\left[\frac{\phi(z_0)}{(z-z_0)^m}(z-z_0)^m + \frac{\phi'(z_0)}{(z-z_0)^{m-1}}(z-z_0)^m + \dots + \frac{\phi^{m-1}(z_0)(z-z_0)^m}{(m-1)!\,(z-z_0)}\right]\Bigg|_{z=z_0}$$

Ex :-

1) $f(z) = \dfrac{(z+2)^3}{(1-z)}$ Residue at $z=1$.

$\Rightarrow \operatorname{Res} = \lim\limits_{z\to 1}\left[(z-1)^1\,\dfrac{(z+2)^3}{1-z}\right]$.

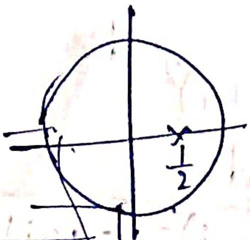
$= -27$.

2) $f(z) = \dfrac{z^2+1}{(z-2)^2}$ at $z=2$.

$\operatorname{Res} f(z) = \dfrac{d^{2-1}}{dz^{2-1}}\left[(z-2)^2 f(z)\right]\Big|_{z=2}$

$= \dfrac{d}{dz}(z^2+1)\Big|_{z=2} = 2z\Big|_{z=2} = 4$

3) $\oint \dfrac{z^2\,dz}{\left(z-\frac{1}{2}\right)\left(z+\frac{7}{2}\right)\left(z-\frac{5}{2}\right)}$ $\quad C: |z|=1$.



$= 2\pi i \operatorname{Res} f(z)\Big|_{z=\frac{1}{2}}$

$\operatorname{Res} f(z) = \lim\limits_{z\to\frac{1}{2}}\left[\left(z-\frac{1}{2}\right)\times\dfrac{z^2}{\left(z-\frac{1}{2}\right)\left(z+\frac{7}{2}\right)\left(z-\frac{5}{2}\right)}\right]$

$= \dfrac{\frac{1}{4}}{4\times -2} = -\dfrac{1}{32}$.

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· Examples :-

1) $\oint \frac{z^3}{(z+1)^3}\,dz = -6\pi i$ $\xrightarrow{\text{By this formula}}$ $\boxed{f^n(z) = \frac{n!}{2\pi i}\oint_c \frac{f(z')\,dz'}{(z'-z)^{n+1}} \quad \text{when } z \text{ inside } C}$

$C: |z| = 3.$

$= \oint_c g(z)\,dz$

$z=-1$, 3rd order pole $n=3$.

$= 2\pi i \sum_i \operatorname{Res} f(z_i).$

$= 2\pi i \left[\frac{1}{(3-1)!}\frac{d^{3-1}}{dz^{3-1}}\left[(z-(-1))^3 \frac{z^3}{(z+1)^3}\right]\right]\Bigg|_{z=-1}.$

$= 2\pi i \,\frac{1}{2}\,\frac{d^2}{dz^2}(z^3)\Big|_{z=-1}.$

$= 2\pi i \, 6z\Big|_{z=-1} = -6\pi i.$

2) $f(z) = \frac{\sin z}{z}$

$= \frac{1}{z}\left(z - \frac{z^3}{3!} + \cdots\right)$

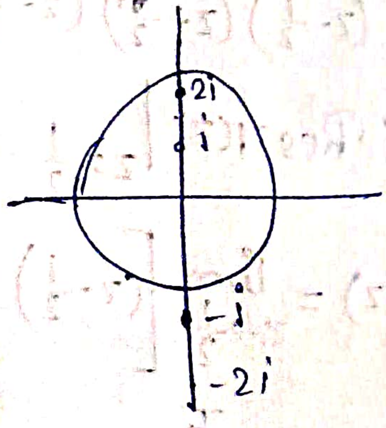
$= 1 - \frac{z^2}{3!} + \cdots$

3) $I = \oint_c \frac{1}{(z^2+4)^2}\,dz$

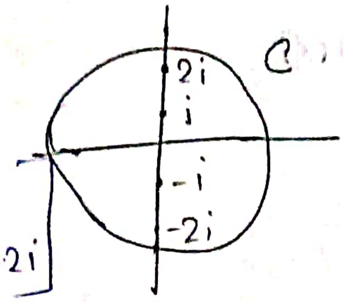
$= \oint \frac{dz}{(z-2i)^2 (z+2i)^2}$

$= 2\pi i \,\frac{d}{dz}\left[\frac{1}{(z+2i)^2}\right]\Bigg|_{z=2i}$

$= \frac{\pi}{16}.$



4. $\oint_C \frac{z}{z^2+4}\,dz$



$= 2\pi i\left[\mathrm{Res}\big|_{z=2i} + \mathrm{Res}\big|_{z=-2i}\right]$

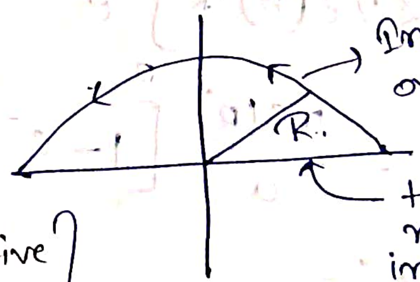
Calculate it.

$=$

Jordon's Lemma

$$J(R,a) = \int_{\Gamma_R} e^{iaz} g(z)\,dz \quad \text{with } a>0.$$

Γ_R = Semicircle (Not closed) of radius R in the upper half of the complex plane.



(a) if $g(z)$ continuous

(b) $g(z) = g(Re^{i\theta}) \to 0$ as $R\to\infty$.

$|g(Re^{i\theta})| < \varepsilon(R)$ for ε positive

and $\varepsilon(R) \to 0$ as $R\to\infty$.

does not depend on θ.

$\Rightarrow g(z)$ uniformly goes to zero in $R\to\infty$.

then $\lim_{R\to\infty} J(R,a) \longrightarrow 0$.

• Toy Case :-

$g(z)=1 \Rightarrow J_R = \int e^{iza}\,dz \quad a>0$

$= \int e^{ia(R\cos\theta + iR\sin\theta)}\,dz$

$= \int (e^{iaR\cos\theta}\, e^{-aR\sin\theta} R\,d\theta) ie^{i\theta} \sim Re^{-aR}$.

$R\to\infty$ $\sin\theta$ positive $0<\theta<\pi$.

$\cancel{Re^{-aR}}$

$Re^{-aR} \to 0$ as $R \to \infty$.

Proof :-

$z = Re^{i\theta}$.

$$J(R,a) = iR\int_0^{\pi} g(Re^{i\theta})\, e^{i\theta} e^{iaR\cos\theta - aR\sin\theta}\, d\theta.$$

$|J(R,a)| \leq$ integral of modulus

$$= \int_0^{\pi} R|g| e^{-aR\sin\theta} d\theta.$$

$$\leq \varepsilon(R) R \int_0^{\pi} d\theta\, e^{-aR\sin\theta}$$

$$= 2\varepsilon(R) R \int_0^{\frac{\pi}{2}} d\theta\, e^{-aR\sin\theta}.$$

$0 < \theta < \frac{\pi}{2} \Rightarrow \sin\theta \geq \frac{2\theta}{\pi}$. (Check).

$$e^{-aR\sin\theta} \leq e^{-aR\frac{2\theta}{\pi}}$$

$$|J| \leq 2\varepsilon(R) R \int_0^{\frac{\pi}{2}} e^{\left(-\frac{2aR\theta}{\pi}\right)}$$

$$= \frac{\pi \varepsilon(R)}{a}\left[1 - e^{-aR}\right] \quad R \to \infty \quad J \to 0.$$

Example :-

1) $\displaystyle\int_{\Gamma_R} \frac{e^{iz}}{z^2+1} dz \quad a = 1 > 0$

$$|f| = \frac{1}{|z^2+1|} \leq \frac{1}{|z|^2-1} = \frac{1}{R^2-1}.$$

$$\varepsilon(R) = \frac{1}{R^2-1}.$$

$f(z) \to 0$ as $R \to \infty$.

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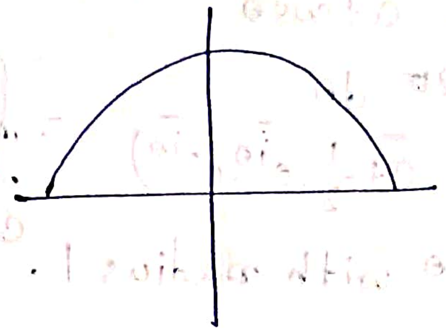
Jordon's Lemma

$$\lim_{R\to a} \int_{\Gamma_R(UHC)} e^{iaz} g(z)\,dz \longrightarrow 0, \quad a>0.$$

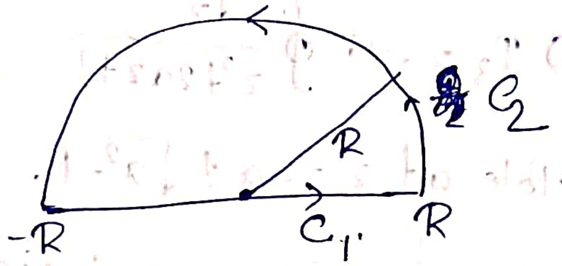
if $g(z) \to 0$ as $R\to\infty$.

1) $$I_1 = \int_{-\infty}^{\infty} \frac{dx}{x^2+1}$$



$$= \int_0^{\infty} \frac{2dx}{x^2+1}.$$

$$= \lim_{R\to\infty} \int_{-R}^{R} \frac{dx}{x^2+1}.$$



$$= \lim_{R\to\infty} \left[\oint_C \frac{dz}{z^2+1} - \int_{C_2} \frac{dz}{z^2+1} \right]$$

$C = C_1 + C_2$.

$\downarrow$

Jordon's Lemma with $a=0$.

For $a=0$, $|zg(z)| \to 0$ as $R\to\infty$ $\Big\}$ Integral $=0$.

$$\Rightarrow \int_{C_2} \frac{dz}{z^2+1} = 0$$

$$\Rightarrow I_1 = \lim_{R\to\infty} \oint_C \frac{dz}{z^2+1} = \text{Res}\left(\frac{1}{z^2+1}\right)\Big|_{z=i} = \pi.$$

2) $$I_2 = \int_{-\infty}^{\infty} \frac{\cos x}{x^2+1}\,dx.$$

$$= \text{Re} \int_{-\infty}^{\infty} \frac{e^{ix}}{x^2+1}\,dx.$$

$a=1>0$

$$= \text{Re} \lim_{R\to\infty} \int_{-R}^{R} \frac{e^{ix}}{x^2+1}\,dx = \text{Re}\left[\oint_C \frac{e^{iz}}{z^2+1}\,dz - \int_{\Gamma_{UHC}} \frac{e^{iz}}{z^2+1}\,dz \right]$$

$$= \mathrm{Re} \oint \frac{e^{iz}}{z^2+1} dz - 0$$ Jordan's Lemma

$$= \mathrm{Re} \left[\mathrm{Res} (\ ) \right] \Big|_{z=+i} = \frac{\pi}{e}.$$

3) $$I_3 = \int_0^{2\pi} \frac{d\theta}{a+\cos\theta} \quad a>1$$

$$= \int_0^{2\pi} \frac{d\theta}{a+\frac{1}{2}(e^{i\theta}+e^{-i\theta})} = \oint_{C:|z|=1} \frac{\frac{dz}{iz}}{a+\frac{1}{2}(z+\frac{1}{z})}$$

$z = e^{i\theta}$ with radius 1.

$$\Rightarrow I_3 = -2i \oint \frac{dz}{z^2+2az+1}$$

Pole at $z = -a \pm \sqrt{a^2-1}$.

<u>Check</u>:- $r_1 = \sqrt{a^2-1} - a$ inside $|z|=1$.

$$\Rightarrow I_3 = 2\pi i \, \mathrm{Res}\Big|_{r_1} = \frac{2\pi}{\sqrt{a^2-1}}.$$

4) $$I_4 = \int_{-\infty}^{\infty} e^{-bx^2} e^{iax} dx \quad b>0.$$

01/09/2025

$$I = \int_{-\infty}^{\infty} e^{-bx^2 + iax}\, dx. \quad (b>0).$$

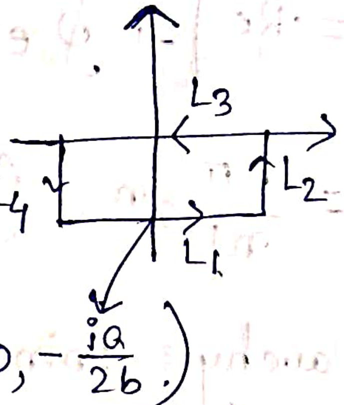
F.T. of a Gaussian

$$-bx^2 + iax = -b\left(x - \frac{ia}{2b}\right)^2 - \frac{a^2}{4b}.$$

$$z = x - \frac{ia}{2b}.$$

$$I = e^{-\frac{a^2}{4b}} \int_{-\infty - \frac{ia}{2b}}^{\infty - \frac{ia}{2b}} e^{-bz^2}\, dz. = e^{-\frac{a^2}{4b}} K.$$

$$C' = L_1 + L_2 + L_3 + L_4$$



$$\int_{C'} e^{-bz^2}\, dz = 0$$

$$\Rightarrow \int_{x=R,\, y=0 \text{ (along } L_3)}^{x=-R,\, y=0} e^{-bz^2} dz + \int_{y=0 \text{ (along } L_4)}^{y=-\frac{a}{2b}} e^{-bz^2} dz + \int_{x=-R \text{ (along } 4)}^{x=R} e^{-bz^2} dz + \int_{y=-\frac{a}{2b} \text{ (along } L_2)}^{y=0} e^{-bz^2} dz = 0$$

$$\Rightarrow -\int_{x=-R,\, y=0 \; (L_3)}^{x=R,\, y=0} e^{-bz^2} dz + \int_{x=-R,\, y=-\frac{a}{2b} \; (L_1)}^{x=R,\, y=-\frac{a}{2b}} e^{-bz^2} dz = L_4 + L_2 \leq 2e^{-R^2} \int_{y=0}^{y=-\frac{a}{2b}} e^{-by^2} dy.$$

$R \to \infty$, R.H.S. $\to 0$

$$\lim_{R\to\infty} \int_{x=-R,\, y=-\frac{a}{2b}}^{x=R,\, y=-\frac{a}{2b}} e^{-bz^2} dz = \int_{x=-R,\, y=0}^{x=R,\, y=0} e^{-bz^2} dz = \sqrt{\frac{\pi}{b}}$$

$$\Rightarrow \int_{-\infty - \frac{ia}{2b}}^{\infty + \frac{ia}{2b}} e^{-bx^2 + iax}\, dx = \sqrt{\frac{\pi}{b}}\, e^{-\frac{a^2}{4b}}.$$

Ex :-

1) $\int_0^{2\pi} d\theta\, e^{a\cos\theta} \cos(a\sin\theta - n\theta)$.

$= \mathrm{Re} \int_0^{2\pi} d\theta\, e^{a\cos\theta}\, e^{i(a\sin\theta - n\theta)}$

$= \mathrm{Re} \int_0^{2\pi} e^{a(e^{i\theta})}\, e^{-in\theta}\, d\theta$.

$z = e^{i\theta}$.

$= \mathrm{Re}\left[\int_{C:|z|=1} e^{az} \cdot \frac{1}{z^n} \cdot \frac{1}{iz}\, dz \right]$

$= \mathrm{Re}\left[-i \oint e^{az} \cdot \frac{1}{z^{n+1}}\, dz \right]$.

$= \frac{2\pi}{n!} a^n$ (By residue theorem).

- Cauchy's Principle value of integral :-

$a \le x \le b$.

$f(x)$ continuous between a and b except at c (finite number of points)

$$\int_a^b f(x)\,dx \overset{\text{Define}}{=\!=} \lim_{\substack{\varepsilon_1 \to 0 \\ \varepsilon_2 \to 0}} \left(\int_a^{c-\varepsilon_1} f(x)\,dx + \int_{c+\varepsilon_2}^{b} f(x)\,dx \right).$$

if the double limit exists.

Comment :-

In some cases limit exists for $\varepsilon_1 = \varepsilon_2 = \varepsilon \to 0$ and does not exist for $\varepsilon_1 \neq \varepsilon_2$.

$$P\int_a^b f(x)\,dx = \lim_{\varepsilon\to 0} \int_a^{c-\varepsilon} f(x)\,dx + \int_{c+\varepsilon}^{b} f(x)\,dx.$$

Example :-

1) $$I = \int_{-1}^{2} \frac{dx}{x^5}.$$

$$= \lim_{\substack{\varepsilon_1\to 0^+ \\ \varepsilon_2\to 0^+}} \int_{-1}^{\varepsilon_1} \frac{dx}{x^5} + \int_{\varepsilon_2}^{2} \frac{dx}{x^5}.$$

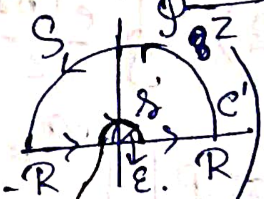
$$= -\frac{1}{4x^4}\Bigg|_{x=-1}^{x=2} + \lim_{\substack{\varepsilon_1\to 0^+ \\ \varepsilon_2\to 0^+}} \left[-\frac{1}{4\varepsilon_1^4} + \frac{1}{4\varepsilon_2^4}\right].$$

limit exist for $\varepsilon_1 = \varepsilon_2 \to 0$

$$P\int_{-1}^{2} \frac{dx}{x^5} = \frac{15}{64}.$$

2) $$I = \int_{-\infty}^{\infty} \frac{e^{ibx}}{x}\,dx.$$

$$\oint \frac{e^{ibz}}{z}\,dz$$



$$= \lim_{\substack{R\to\infty \\ \varepsilon\to 0}} \left[\int_{-R}^{\varepsilon} \frac{e^{ibx}}{x}\,dx + \int_{\varepsilon}^{R} \frac{e^{ibx}}{x}\,dx\right]$$

0 outside the contour C'

Contour integration at z=0 pole. But z=0 on the contour.

$$\Rightarrow \oint_{C'} \frac{e^{ibz}}{z}\,dz = 0 = \int_{-R}^{\varepsilon} \frac{e^{ibx}}{x}\,dx + \int_{\varepsilon}^{R} \frac{e^{ibx}}{x}\,dx + \int_{\delta} dz\, \frac{e^{ibz}}{z} + \int_{S} \frac{e^{ibz}}{z}\,dz$$

$$= P\left[\int_{-R}^{R} \frac{e^{ibx}}{x}\, dx\right] + \int_{s} + \int_{S} .$$

$$z = \varepsilon e^{i\theta} \Rightarrow \frac{dz}{z} = i\, d\theta .$$

$$\lim_{\varepsilon \to 0} \int_{s} dz\, \frac{e^{ibz}}{z} = \lim_{\varepsilon \to 0} \int_{\pi}^{0} d\theta\, e^{(ib\varepsilon e^{i\theta})} = -i\pi .$$

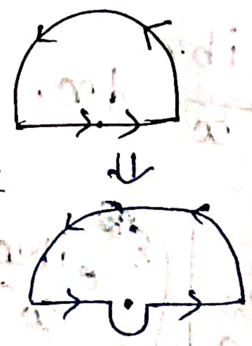
$$\lim_{R \to \infty} \int_{S} dz\, \frac{e^{ibz}}{z} \longrightarrow 0 .$$

$$\Rightarrow 0 = P \int_{-\infty}^{\infty} dx\, \frac{e^{ibx}}{x} - i\pi + 0 .$$

$$\Rightarrow P \int_{-\infty}^{\infty} dx\, \frac{e^{ibx}}{x} = i\pi .$$

$$P \int \frac{\sin bx}{x}\, dx = \pi$$ (Equating imaginary part).

2) $\oint \frac{e^{ibz}}{z}\, dz$ along



$$= P\left[\int_{-\infty}^{\infty} \frac{e^{ibx}}{x}\, dx\right] + \underbrace{S}_{0} + \underbrace{\int dz\, \frac{e^{ibz}}{z}}_{+i\int_{\pi}^{2\pi} d\theta .}$$

$$= P[\ \ ] + \pi i = 2\pi i\, \mathrm{Res}(f(z))\big|_{z=a}$$

$$= 2\pi i$$

$$\Rightarrow P[\ \ ] = 2\pi i - \pi i = \pi i .$$

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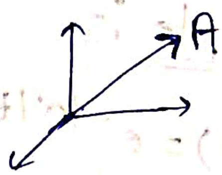
Fourier Transformation

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi x}{L} + b_n \sin\frac{n\pi x}{L}\right).$$

$$a_n = \frac{1}{L}\int_{-L}^{L} f(x)\cos\left(\frac{n\pi x}{L}\right)dx.$$



$$b_n = \frac{1}{L}\int_{-L}^{L} f(x)\sin\left(\frac{n\pi x}{L}\right)dx$$

$$f(x) = \frac{1}{2L}\int_{-L}^{L} f(x')\,dx' + \sum_{n=1}^{\infty}\frac{1}{L}\cos\left(\frac{n\pi x}{L}\right)\int_{-L}^{L} f(x')\cos\left(\frac{n\pi x'}{L}\right)dx'$$
$$+ \sum_{n=1}^{\infty}\frac{1}{L}\sin\left(\frac{n\pi x}{L}\right)\int_{-L}^{L} f(x')\sin\left(\frac{n\pi x'}{L}\right)dx.$$

$$\cos(A+B) = \cos A\cos B - \sin A\sin B.$$

$$f(x) = \frac{1}{2L}\int f(x')\,dx' + \frac{1}{L}\sum_{n=1}^{\infty}\int_{-L}^{L} f(x')\cos\left(\frac{n\pi}{L}(x-x')\right)dx'$$

$\frac{n\pi}{L} = k$, $\Delta k = \frac{\pi}{L}\Delta n = \frac{\pi}{L}$.

$L\to\infty$, 1st term drops. $\int f(x')\,dx'$ finite.

$$f(x) = \frac{1}{\pi}\int_0^{\infty} dk\int_{-\infty}^{\infty} f(x')\cos(k(x-x'))\,dx'.$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} dk\int_{-\infty}^{\infty} f(x')\cos(k(x-x'))\,dx'$$

$$\frac{1}{2\pi}\int_{-\infty}^{\infty} dk\int f(x')\sin k(x-x')\,dx' = 0$$

$$f(x) = \frac{1}{2\pi}\int dk\int f(x')\,e^{ik(x-x')}\,dx'$$

$$f(x) = \frac{1}{\sqrt{2\pi}}\int dk\,e^{ikx}\left[\frac{1}{\sqrt{2\pi}} f(x')\,e^{-ikx'}dx'\right].$$

$$\Rightarrow f(x) = \frac{1}{\sqrt{2\pi}}\int dk\,e^{ikx}\,g(k).$$

$$g(k) = \mathcal{F}(k) = \frac{1}{\sqrt{2\pi}} \int f(x') e^{-ikx'} dx'$$

$\longrightarrow$ Fourier transform of $f(x)$.

Example :-

1) $f(t) = e^{-|\alpha| t}$ $\alpha > 0$.

$$\Rightarrow g(\omega) = \mathcal{F}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{0} e^{\alpha t + i\omega t} + \frac{1}{\sqrt{2\pi}} \int_{0}^{\infty} e^{-\alpha t + i\omega t} dt.$$

$$= \frac{1}{\sqrt{2\pi}} \left[\frac{1}{\alpha + i\omega} + \frac{1}{\alpha - i\omega} \right] = \frac{1}{\sqrt{2\pi}} \frac{2\alpha}{\alpha^2 + \omega^2}.$$

2) $f(t) = \delta(t)$

$$\Rightarrow g(\omega) = \frac{1}{\sqrt{2\pi}} \int \delta(t) e^{i\omega t} = \frac{1}{\sqrt{2\pi}}$$

3) $f(t) = \frac{1}{\sqrt{2\pi}} \frac{2\alpha}{\alpha^2 + t^2}$

$$\Rightarrow g(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\alpha}{\alpha^2 + t^2} e^{i\omega t} dt.$$

$$= \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{2\alpha}{(\alpha + it)(\alpha - it)} e^{i\omega t} dt.$$

when $\omega > 0 \longrightarrow g(\omega) = \frac{1}{2\pi} (2\pi i) \operatorname{Res}\Big|_{i\alpha} = e^{-\alpha\omega}$

when $\omega < 0 \longrightarrow g(\omega) = \frac{1}{2\pi} (-2\pi i) \operatorname{Res}\Big|_{-i\alpha} = e^{\alpha\omega}$.

(Check by explicit calculation)

$$g(\omega) = e^{-\alpha|\omega|}$$

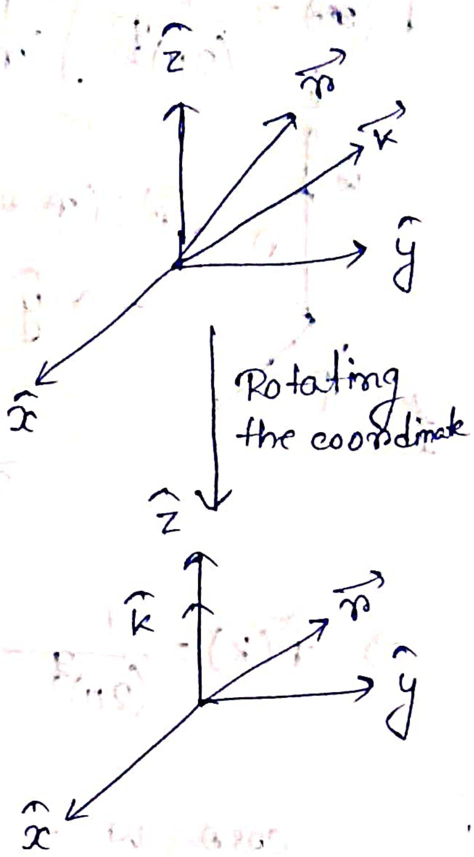
Fourier Transform of Yukawa Potential

$$V(\vec{r})\,\cancel{f(\vec{x})} = \frac{1}{|\vec{r}|}e^{-b|\vec{r}|}. \qquad b>0$$

$$= f(\vec{x}') \qquad 0<|\vec{r}|<\infty.$$

3-D F.T.

$$F(\vec{k}) = \frac{1}{(2\pi)^{3/2}}\int \frac{1}{r}\, e^{-i\vec{k}\cdot\vec{r} - br}\, d^3r$$



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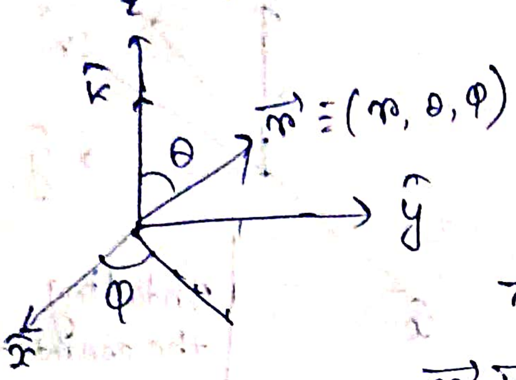
$$f(\vec{x}) = \frac{1}{|\vec{r}|} e^{-b|\vec{r}|} \qquad b > 0 \qquad 0 < r < \infty.$$

$$\tilde{f}(k) = \frac{1}{(2\pi)^{\frac{3}{2}}} \int \frac{1}{r} e^{-i\vec{k}\cdot\vec{r} - br} d^3r$$



$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}.$$

$$\vec{r}\cdot\vec{k} = |\vec{r}|\,|\vec{k}| \cos\theta.$$

$$\Rightarrow \tilde{f}(k) = \frac{1}{(2\pi)^{\frac{3}{2}}} \int \frac{1}{r} e^{-i|\vec{k}||\vec{r}|\cos\theta - br} r^2 \sin\theta \, dr\, d\theta\, d\varphi.$$

$$\cos\theta = w \longrightarrow \theta = 0 \Rightarrow w = 1$$
$$\theta = \pi \Rightarrow w = -1.$$

$$d(\cos\theta) = -\sin\theta \, d\theta.$$

<u>θ part :-</u>

$$\int_0^{\pi} d\theta \, e^{-ikrw} \sin\theta = -\int_1^{-1} dw \, e^{-ikrw} = \int_{-1}^{1} dw \, e^{-ikrw}$$

$$= -\frac{1}{ikr}\left[e^{-ikrw}\right]_{-1}^{1}$$

$$= -\frac{1}{ikr}\left[e^{-ikr} - e^{ikr}\right].$$

$$= -\frac{1}{ikr}\left[\cos kr - i\sin kr - \cos kr - i\sin kr\right]$$

$$= \frac{2\sin kr}{kr}.$$

$$\mathcal{F}(k) = \frac{12\pi}{(2\pi)^{\frac{3}{2}}} \frac{2}{k} \int_0^{\infty} dr\, e^{-br} \sin kr$$

$$\int_0^{\infty} e^{-br} \sin kr\, dr = \operatorname{Im} \int_0^{\infty} e^{-r(b-ik)} dr$$

$$= \operatorname{Im} \left[\frac{1}{ik-b} e^{-r(b-ik)} \Big|_0^{\infty} \right]$$

$$= \operatorname{Im} \left(\frac{b+ik}{b^2+k^2} \right) = \frac{k}{b^2+k^2}$$

$$\Rightarrow \mathcal{F}(k) = \frac{4\pi}{(2\pi)^{\frac{3}{2}}} \left(\frac{1}{k^2+b^2} \right)$$

- $f \xleftrightarrow{FT} g$

$$\left.\begin{aligned} g'(k) &= \frac{1}{\sqrt{2\pi}} (-ik)\, g(k) \\ g''(k) &= -k^2 g(k) \end{aligned}\right\} \text{Subject to Boundary conditions.}$$

- Fourier Transform of even (odd) function is also even (odd).

- $$\int_{-\infty}^{\infty} \tilde{f}^*(k)\, \tilde{g}(k)\, dk = \int_{-\infty}^{\infty} f^*(x)\, g(x)\, dx.$$

$$\int_{-\infty}^{\infty} |\mathcal{F}(k)|^2 dk = \int_{-\infty}^{\infty} |f(x)|^2 dx$$

Convolution Theorem

$$(f*g)(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x-y)\, g(y)\, dy.$$

1. $f*g = g*f$
2. If f and g are integrable then so is $(f*g)(x)$.

$$\boxed{\text{F.T.}\left[(f*g)\right] = \tilde{f}(k)\,\tilde{g}(k).}$$

Proof :-

$$(f*g)(x) = h(x)$$

$$\tilde{h}(k) = \frac{1}{2\pi}\int_{-\infty}^{\infty} dx\, e^{-ikx}\int_{-\infty}^{\infty} f(x-y)\, g(y)\, dy.$$

$$= \frac{1}{2\pi}\int_{-\infty}^{\infty} dx\, e^{-ikx}\int_{-\infty}^{\infty} f(x-y)\, g(y)\, e^{-iky} e^{iky}\, dy$$

$$= \frac{1}{\sqrt{2\pi}}\int_{-\infty}^{\infty} g(y)\, e^{-iky} dy \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} f(x-y)\, e^{-ik(x-y)}\, dx.$$

$$= \frac{1}{\sqrt{2\pi}}\int g(y)\, e^{-iky}\, dy \times \frac{1}{\sqrt{2\pi}}\int f(z)\, e^{-ikz}\, dz$$

$$= \tilde{g}(k)\,\tilde{f}(k).$$

$$\left.\begin{array}{l} D f(x) = g(x) \\ f \sim \tilde{f}(k)\, e^{-ikx} \\ g \sim \tilde{g}(k)\, e^{-ikx}. \end{array}\right] \Longrightarrow$$

Algebraic expression
$\tilde{f}(k)$ and $\tilde{g}(k)$
$\Downarrow$
$f(x) = \int \tilde{f}(k)\, e^{-ikx}.$

$$-\frac{d^2 y}{dx^2} + \alpha^2 y = f(x) \qquad \alpha > 0$$

$$f(x) = \frac{1}{(2\pi)^{\frac{1}{2}}}\int \tilde{f}(k)\, e^{ikx}\, dk.$$

$$y = \frac{1}{(2\pi)^{\frac{1}{2}}}\int \tilde{y}\, e^{ikx}\, dk$$

$$\Rightarrow (-k^2)\,\tilde{y}(k) + \alpha^2\,\tilde{y}(k) = \tilde{f}(k)$$

$$\Rightarrow \tilde{y}(k) = \frac{\tilde{f}(k)}{k^2+\alpha^2} \longrightarrow \text{Now algebraic expression}$$

$$y(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{\tilde{f}}{k^2+\alpha^2}\, e^{ikx}\, dk.$$

- $$G_\alpha(x) = \frac{1}{2\alpha}\, e^{-\alpha|x|}$$

$$\tilde{G}_\alpha(k) = \frac{1}{k^2+\alpha^2}$$

$$Y(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \tilde{G}_\alpha(k)\, \tilde{f}(k)\, e^{ikx}\, dk.$$

$$= (G_\alpha * f)(x).$$

$$= \int_{-\infty}^{\infty} G_\alpha(x-\xi)\, f(\xi)\, d\xi$$

$$= \int_{-\infty}^{\infty} \frac{1}{2\alpha}\, e^{-\alpha|x-\xi|}\, f(\xi)\, d\xi.$$